MALFUNCTION INDICATOR LAMP (MIL)—DIAGNOSTIC TROUBLE CODES (DTCS) P0442 AND P0455—NO DRIVEABILITY CONCERNS	Article No. 99-23-4

FORD: 1997-2000 CROWN VICTORIA, TAURUS
1998-1999 ESCORT
1998-2000 CONTOUR, MUSTANG
2000 FOCUS
1996-2000 EXPLORER
1997 F-250 LD
1997-2000 E-150, EXPEDITION, F-150
1998-2000 RANGER, WINDSTAR
1999-2000 F-250

LINCOLN: 1997-2000 TOWN CAR
1998 MARK VIII
1998-2000 CONTINENTAL
2000 LS
1998-2000 NAVIGATOR

MERCURY: 1997-2000 GRAND MARQUIS, SABLE
1998-1999 TRACER
1998-2000 MYSTIQUE
1999-2000 COUGAR
1997-2000 MOUNTAINEER

This TSB article is being republished in its entirety to include vehicles built through the 2000 model year and to revise the following NOTE.

NOTE
FOR 1998 MODEL YEAR MUSTANG, THIS TSB IS ONLY APPLICABLE TO CALIFORNIA VEHICLES.

ISSUE

There may be difficulty diagnosing a vehicle with Malfunction Indicator Lamp (MIL) illuminated and Diagnostic Trouble Codes (DTCs) P0442 and P0455 stored in continuous memory with no driveability concerns present.

ACTION

Diagnose DTCs P0442 and P0455 by referring to the following Evaporative Emission (EVAP) Running Loss System Monitor Procedure, Service Bay Diagnostic System (SBDS) Static Leak Test, and EVAP Running Loss System Drive Cycle for details.

EVAP RUNNING LOSS SYSTEM MONITOR PROCEDURE

This is an on-board strategy designed to detect a leak from a hole (opening) equal to or greater than 1.016mm (0.040″) in the EVAP system. The proper function of the individual components of the EVAP system as well as its ability to flow fuel vapor to the engine is also examined.

The EVAP Running Loss System Monitor relies on the individual components of the EVAP system to apply vacuum to the fuel tank and then seal the system. The fuel tank pressure is then monitored to determine the total vacuum lost (bleed-up) for a calibrated period of time. Inputs from the Engine Coolant Temperature (ECT) sensor or Cylinder Head Temperature (CHT) sensor, Intake Air Temperature (IAT) sensor, Mass Air Flow (MAF) sensor, Vehicle Speed Sensor (VSS) or Output Shaft Sensor (OSS), Fuel Level Input (FLI) sensor, and Fuel Tank Pressure (FTP) sensor are required to enable the EVAP Running Loss System Monitor. The Powertrain Control Module (PCM) monitors the time since power down and will not allow the EVAP Running Loss System Monitor to run until the minimum soak timer (6-10 hours) has expired.

NOTE
DURING THE EVAP RUNNING LOSS SYSTEM MONITOR DRIVE CYCLE, A PCM RESET (CLEARING CONTINUOUS CODES) WITH KEY ON, ENGINE OFF (KOEO) WILL BYPASS THE MINIMUM SOAK TIME REQUIRED TO COMPLETE THE MONITOR. THE EVAP RUNNING LOSS SYSTEM MONITOR WILL NOT RUN IF THE KEY IS TURNED OFF AFTER A PCM RESET UNLESS THE MINIMUM SOAK TIME WAS REACHED. THE EVAP RUNNING LOSS SYSTEM MONITOR WILL NOT RUN IF A MAF, ECT, IAT, VSS, FLI, OR FTP SENSOR FAILURE(S) IS INDICATED. THE EVAP RUNNING LOSS SYSTEM MONITOR WILL NOT INITIATE UNTIL THE HEATED OXYGEN SENSOR (HO2S) MONITOR HAS COMPLETED.

The EVAP Running Loss System Monitor is executed by the individual components of the EVAP Running Loss System as follows:

1. The EVAP canister purge valve applies a vacuum to the fuel tank. A minimum duty cycle on the EVAP canister purge valve (75%) must be met before the EVAP Running Loss System Monitor can begin.
2. The Canister Vent (CV) solenoid will close (100% duty cycle) to seal the EVAP system.
3. The FTP sensor will be used by the EVAP Running Loss System Monitor to determine if the target vacuum on the fuel tank is being reached to perform the leak check.
4. If the initial target vacuum cannot be reached, DTC P0455 (large leak detected) will be set. The EVAP Running Loss System Monitor will abort and not continue with the leak check portion of the test.

Once the target vacuum is obtained on the fuel tank, the EVAP canister purge valve closes and the change in the fuel tank vacuum will be monitored over a calibrated period of time. If the change in fuel tank vacuum is less than the calibrated threshold, the EVAP Running Loss System Monitor passes.

If the change in fuel tank vacuum exceeds the calibrated threshold, the test will rerun up to three (3) times. If the calibrated threshold is still being exceeded after three tests, a vapor generation check must be performed before DTC P0442 (small leak detected) will be set. The system accomplishes this by returning the EVAP system to atmospheric pressure by closing the EVAP canister purge valve and opening the CV solenoid.

Once the FTP sensor observes the fuel tank is at atmospheric pressure (approximately 2.6 volts), the CV solenoid closes (seals the EVAP system). If the fuel tank pressure buildup exceeds the calibrated threshold, the leak test results are invalidated due to excessive vapor (the EVAP Running Loss System Monitor will pass and complete). If the fuel tank pressure buildup does not exceed the calibrated threshold, the leak test results are valid and DTC P0442 will be set.

5. The MIL is activated for DTCs P0442 and P0455 after two (2) occurrences of the same fault.

NOTE
REFER TO THE LATEST POWERTRAIN CONTROL/EMISSIONS DIAGNOSIS (PC/ED) SERVICE MANUAL ON SBDS OR SERVICE BAY TECHNICAL SYSTEM (SBTS) FOR INFORMATION ON DTCs P0443, P0446, P0452, P0453, P1450, AND P1451.

SBDS STATIC LEAK TEST

1. Perform SBDS startup.
2. Select "Powertrain" from the Main Menu.
3. Enter the Vehicle Identification Number (VIN) and then press "Continue."

4. Select "Electronic Engine Control" under Categories and "DCL Display" under Tools, then press "Continue."
5. If applicable, select the appropriate vehicle qualifier and then press "Continue."
6. Follow the instructions on the connection screen, make sure vehicle is in Park or Neutral, chock the tires, start vehicle, and press "Continue."
7. Select "Table."
8. Select EVM (units of mode and %), ECV (units of mode), FTP (units of volts), IAC (units of %), RPM, and then press "Continue."

NOTE
IF ALL OF THE PARAMETERS FOR STEP 8 ARE NOT AVAILABLE USING THE LATEST SBDS CD-ROM RELEASE, RETURN TO THE MAIN MENU AND USE THE FOLLOWING VIN FOR EVAP TESTING ONLY: XXXXM91V6VXXXXXXX.

9. Use the pendant on the Portable Vehicle Analyzer (PVA) to select the ECV (on/off) using the "#" button and then press "Yes." The ECV should now have an "=" sign next to it. Pressing the "Yes" button turns the ECV on and off. Turn the ECV on (shuts off venting of system).
10. Repeat the procedure in Step 9 to turn the EVM on (opens system to engine vacuum). The FTP voltage should drop to 0.8 volts or less (voltage should be attained in approximately 1 minute). If the FTP voltage does not reach 0.8 volts or less, then a large leak has been detected (follow the latest PC/ED Service Manual diagnostics contained on SBDS or SBTS).

NOTE
IF THE VEHICLE STALLS, IT WILL ALSO BE NECESSARY TO RAISE THE ENGINE SPEED TO 1500 RPM. THIS CAN BE ACCOMPLISHED BY USING THE PENDANT ON THE PVA TO INCREASE THE IAC (%) TO RAISE THE RPM.

11. Use the pendant on the PVA to turn the EVM off (holds system pressure constant), FTP voltage should stabilize within 20-30 seconds and then hold at a constant voltage. If the FTP voltage changes more than 0.4 volts in 1 minute, then a small leak has been detected (follow the latest PC/ED Service Manual diagnostics contained on SBDS or SBTS).

NOTE
COMMANDING THE ECV ON FOR MORE THAN 5 MINUTES MAY CAUSE DAMAGE TO THE CANISTER VENT SOLENOID.

EVAP RUNNING LOSS SYSTEM DRIVE CYCLE

WARNING
STRICT OBSERVANCE OF POSTED SPEED LIMITS AND ATTENTION TO DRIVING CONDITIONS ARE MANDATORY WHEN PROCEEDING WITH THE FOLLOWING DRIVE CYCLE.

1. Attach a scan tool and perform a PCM reset (clear continuous codes) to bypass the minimum soak timer. Leave the key in the On position and start the vehicle. Once started, the engine must not be turned off. If the engine is turned off, it will be necessary to perform a PCM reset and start over.
2. Use the scan tool to access the ECT, FLI, IAT, FTP V, EVAPPDC, EVAPCV, VSS, and DTC CNT Parameter Identification Displays (PIDs). Warm the vehicle until the ECT reaches a minimum of 54°C (130°F).
3. Remove the gas cap and verify that the FTP voltage reads between 2.4 and 2.8 volts. If voltage is not in this range, refer to the latest PC/ED Service Manual diagnostics contained on SBDS or SBTS for FTP diagnostics (Pinpoint Test Step HX22); otherwise, reinstall the gas cap and proceed to Step 4.
4. Drive the vehicle on a smooth road at a constant speed between 64 and 105 km/h (40 and 65 mph) with the throttle as steady as possible. During this time, verify the following EVAP monitor entry conditions:
   - FLI between 15 and 85% and stable ±5% (fuel may need to be added or removed)
   - FTP V stable within ±0.1 volts
   - IAT between 10 and 38°C (50 and 100°F)

5. While continuing to drive at a constant speed (between 64 and 105 km/h (40 and 65 mph)), monitor the EVAPPDC, EVAPCV and FTP V PIDs. The EVAPPDC will increase to at least 75% and then return to 0%. EVAPPDC will again increase to at least 75% and then return to 0%. The third time the EVAPPDC increases to at least 75%, the EVAPCV will change from 0% to 100% (CV solenoid closed) to seal the system, the EVAPPDC will then return to 0%, and the leak test will begin to run. At this time, the FTP V should be stable at 1.2-1.6 volts (if the FTP V does not remain stable, then a small leak has been detected).

Continue to drive at steady throttle with light steering until the EVAPCV displays 0%. Monitoring the DTC CNT will help determine if a DTC is set. If this does not occur as described, refer to the following Note; otherwise proceed to Step 6.

NOTE

- **THE EVAP MONITOR LEAK TEST MAY RUN ON THE FIRST OR SECOND TIME THE EVAPPDC INCREASES TO MORE THAN 75% IF THE AMOUNT OF FUEL VAPOR IN THE FUEL TANK THAT HAS BEEN DETECTED BY THE EEC IS LOW.**
- **THE EVAP MONITOR WILL NOT RUN IF THE HO2S MONITOR HAS NOT COMPLETED (VIEW THE ON-BOARD SYSTEM READINESS TESTS TO DETERMINE IF THE MONITOR HAS COMPLETED).**
- **IF THE EVAP MONITOR ENTRY CONDITIONS DESCRIBED IN STEP 4 ARE NOT MAINTAINED, THEN THE EVAP MONITOR WILL NOT CONTINUE TO RUN.**
- **IF THE EVAPPDC STAYS AT 99% AND THE FTP V DOES NOT REACH 1.2-1.6 VOLTS, THEN THE EVAP MONITOR HAS DETECTED A LARGE LEAK (DTC P0455 WILL BE SET).**
- **IF THE FTP V DOES NOT REMAIN STABLE AT 1.2-1.6 VOLTS (INCREASES MORE THAN 0.4 VOLTS) WHILE THE EVAP MONITOR IS RUNNING, THEN THE PCM WILL RETRY THE LEAK TEST A MAXIMUM OF THREE (3) TIMES. IF THE EVAP MONITOR LEAK TEST FAILS ALL THREE TIMES, THEN A VAPOR GENERATION CHECK WILL BE PERFORMED. IF VAPOR GENERATION HAS NOT BEEN DETECTED, THEN THE EEC HAS DETERMINED THERE IS A SMALL LEAK (DTC P0442 WILL BE SET).**
- **IF ANY DTCs ARE SET, FOLLOW THE LATEST PC/ED SERVICE MANUAL DIAGNOSTICS CONTAINED ON SBDS OR SBTS.**

6. Bring the vehicle to a stop.

7. With the scan tool, view the On-Board System Readiness Tests to determine if the EVAP Monitor has been completed. If the EVAP Monitor has completed and there are no continuous DTCs, then the PCM has not detected any concerns. Repeat this Drive Cycle if the EVAP Monitor has not completed and there are no continuous DTCs in memory.

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 99-1-9
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 206000, 698298